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Rear axle with integrated electric motor

Abstract

A truck, comprising a rear axle, comprising a motor housing extending laterally between rear wheels of the truck, and an electric motor and power transmission coaxially aligned in said motor housing and arranged for driving the rear wheels; and a suspension system, for suspending the rear axle to a chassis of the truck, said suspension system at least comprising a pair of springs and shock absorbers, and an anti-roll stabilizer and a guiding rod linkage for allowing movement in a vertical direction relative to the chassis; wherein the motor housing is limited by a minimum vertical ground clearance between the road surface and a bottom surface of the motor housing; wherein the motor housing comprises a reinforced central top section providing a guiding rod mount for pivotally mounting the guiding rod linkage to the rear axle; wherein the motor housing further comprises suspension yokes integrated into the housing and extending on either side of the motor housing near both axial ends, said suspension yokes providing spring mounts and shock absorber mounts for mounting said opposed pair of springs and shock absorbers, wherein the motor housing comprises a rear section outer surface provided with connectors, to provide input ports to the electric motor and wherein said connectors are provided in an orientation to guide one or more high voltage power cables adjacent the rear section outer surface in a vertical upward direction.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is a U.S. National Stage application under 35 U.S.C. § 371 of International Application PCT/NL2021/050702 (published as WO 2022/108444 A1), filed Nov. 16, 2021, which claims the benefit of priority to Application NL 2026916, filed Nov. 17, 2020. Benefit of the filing date of these prior applications is hereby claimed. Each of these prior applications is hereby incorporated by reference in its entirety.

DESCRIPTION OF THE PRIOR ART

(2) The transformation from diesel powered heavy commercial vehicles to electric powered heavy electric vehicles brings many challenges, one of which is the integration of modules for powering an electric vehicle into the conventional truck architecture.

(3) To propel an electric powered vehicle, the electric motor or electric motors are new components to be packaged in the vehicle. Architectural concepts include placing the electric motor and transmission in the chassis, e.g. on the spot where the internal combustion engine is located. In contrast to conventional trucks the driven axle may instead powered by an internal electric motor resulting in a compact design where an electric powertrain can have all of the functions of a conventional powertrain embedded in a single body, also known as an E-axle. Typically, the E-axle receives its energy from an electric power source, e.g. a battery pack or fuel cells, located elsewhere in the truck and connected through one or multiple power cables.

(4) DE102016004367 describes a rigid axle driven by an electric drive having a suspension for an axle with an electric motor and gearbox arranged in alignment with one another. Air bellows supports extend transversely to the driven axle and are attached to a housing of the gearbox.

(5) Publication EP2991889 describes another approach to integrating modules for powering an electric vehicle into a truck architecture. A chassis component is disclosed for carrying a load applied to the vehicle, such as a fifth wheel, which is distributed to the wheels of the vehicle. A hollow space in the chassis component can be used for housing an electrical drive unit and a power transmission. The chassis component may function as a complete suspension component for connecting the wheels to the chassis. However, the increased load characteristics of the chassis component relative to a that of a conventional truck, would require design modifications to the suspension setup.

(6) For configurations in which the E-axle is directly attached to the wheels of the truck, the E-axle moves with respect to the electric power source while the truck is driving. With this concept the conventional rear axle setup and power transmission can be conserved.

(7) However, to allow for relative movement of the electric motor, the power cable should also

bend or deflect with the movement of the E-axle. This deflection can cause a durability problem due to an inherent fatigue limit of the power cable.

(8) Considering various road surface conditions, e.g. bumps and holes, and load conditions of the driven axle, e.g. cornering, accelerating, braking and trailer load distribution, this poses a significant challenge for trucks with E-axles, with respect to power cabling fatigue issues.

(9) There is a need for further improvement with regards to cable supports in trucks, that alleviate these or other problems.

SUMMARY OF THE INVENTION

(10) In one aspect, it is aimed to provide a truck comprising a rear axle and a suspension system. The rear axle comprises a motor housing extending laterally between rear wheels of the truck. The rear axle further comprises an electric motor and power transmission coaxially aligned in said motor housing and arranged for driving the rear wheels. The suspension system suspends the rear axle to a chassis of the truck, and at least comprises a pair of springs and shock absorbers, and an anti-roll stabilizer and a guiding rod linkage for allowing movement in a vertical direction relative to the chassis. The motor housing is limited by a minimum vertical ground clearance between the road surface and a bottom surface of the motor housing. The motor housing comprises a reinforced central top section providing a guiding rod mount for pivotally mounting the guiding rod linkage to the rear axle. The motor housing further comprises suspension yokes integrated into the housing and extending on either side of the motor housing near both axial ends. The suspension yokes provide spring mounts and shock absorber mounts for mounting said opposed pair of springs and shock absorbers. The motor housing comprises a rear section outer surface provided with connectors, to provide input ports to the electric motor. The connectors are provided in an orientation to guide one or more high voltage power cables adjacent the rear section outer surface in a vertical upward direction. For the high voltage cables, the upward vertical orientation will minimize the bending stresses induced by the axle suspension movement.

(11) Said connectors are thus provided with an orientation to guide the one or more high voltage cables in upward direction minimize fatigue stress, since high voltage cable connectors typically are provided from low resistance metals, e.g. copper or copper alloys, that are susceptible to bending stress. The one or more high voltage cables electrically connects the electric power source to the E-axle. To further alleviate bending stress, a support beam may form a mechanical support structure to lead the power cable along a cable path extending between a chassis mount and an E-axle mount. The support beam may comprise one or more bending stiffness elements, forming a stiffness of the support beam that is highest in a middle section and that decreases towards the chassis mount and the E-axle mount, evenly distributing the bending stress in the support beam along the cable path when the E-axle moves relative to the chassis, to further prevent local fatigue failure of the power cable.

(12) In some embodiments, the support beam may extend from a chassis mount in a backward direction along a longitudinal member of the chassis and connects the E-axle mount along a vertical direction, to said a rear section outer surface provided with connectors, to provide input ports to the electric motor.

(13) In some embodiments, an electrically conductive core may be mounted between the chassis mount and the E-axle connector, to make a ground connection between the E-axle and the chassis and the support beam may be mounted to the conductive core along at least a part of the cable path, e.g. to form an electrically insulating sleeve with a bending stiffness that varies along the cable path.

(14) In other or further embodiments, the support beam comprises bending stiffness elements comprising one or more flanges extending along the cable path. The flanges may have a flange height that is highest in the middle section of the support beam and that decreases towards the chassis mount and the E-axle mount. In one embodiment, the bending stiffness elements may comprise two vertically extending flanges forming an H-beam to prevent the support beam from

deflecting in a lateral direction.

(15) In yet further embodiments, the support beam comprises laterally extending cable supports arranged for leading multiple power cables adjacent the cable path, each power cable following the cable path at a lateral offset from the support beam. The multiple power cables may be connected to the E-axle by a common power connector, at a distance beyond the E-axle mount to allow disconnecting the power cables without disassembling the support beam from the E-axle mount.

(16) In some embodiments, on each axial end of the motor housing, an anti-roll mount of an anti-roll stabilizer, an opposing pair of spring mounts and a shock absorber mount lie on a common plane perpendicular to a centerline of the rear wheels of the truck. The anti-roll stabilizer may comprise a torsion shaft. The torsion shaft is mounted to the chassis while allowing an axial rotation at an offset to the rear axle, and has a pair of linkage plates on both ends extending from the torsion shaft towards the anti-roll mounts. The anti-roll stabilizer is configured such that a roll movement of the truck, causing a difference in a vertical displacement of the anti-roll mounts on each axial end of the motor housing relative to the chassis, is counteracted by a torsional stiffness of the torsion shaft.

(17) In other or further embodiments, the guiding rod linkage comprises a pair of guiding rods. The pair of guiding rod is pivotally mounted to the chassis at an offset to the rear axle, and extends towards the reinforced central top section.

(18) In some embodiments, the reinforced central top section of the motor housing comprises a number of ribs. The ribs converge toward the guiding rod mount and have a rib height that is highest near the guiding rod mount and that decreases away from the guiding rod mount.

(19) Additionally, or alternatively, housing extensions are coaxially connected to both axial ends of the motor housing for mounting rear wheel brakes. The housing extensions have a proximal base part that is axially connected to the motor housing, and have a distal part extending away from the motor housing toward the rear wheels with a diameter smaller than the proximal base part.

(20) In some embodiments, the motor housing comprises a rear section outer surface provided with connectors, to provide input ports to the electric motor. In a further embodiment, the connectors are provided in an orientation to guide one or more high voltage power cables adjacent the rear section outer surface in a vertical upward direction.

(21) Additionally or alternatively, the motor housing may be arranged for internally carrying a gearbox and/or inverter for powering different groups of coils of the electric motor.

(22) In some embodiments, the motor housing is made of an electrically conductive material to electromagnetically shield at least the electric motor. Additionally or alternatively, the motor housing is made of a thermally conductive material to passively cool e.g. the electric motor or the power transmission.

(23) In yet further embodiments, the rear axle is provided with an internal cooling circuit for circulating coolant fluid through the axle housing and one or more coolant hose connectors on the rear section outer surface for receiving one or more coolant hoses, to actively cool at least the electric motor. The coolant fluid may additionally be employed as lubricant for lubricating rotating parts inside the motor housing, such as the power transmission.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The invention will be further elucidated in the figures:

(2) FIG. 1 shows a top view of a truck with a rear axle with integrated electric motor according to a first embodiment;

(3) FIG. 2 shows a further embodiment of a truck with one or more high voltage cables;

(4) FIG. 3 shows a perspective view on a support beam for a high voltage cable;

- (5) FIG. 4 provides an isometric bottom view of a further embodiment of the truck;
- (6) FIG. 5 provides an isometric top view of another further embodiment of the truck;
- (7) FIG. 6 provides an exploded view of yet another or further embodiment of the truck.
- (8) FIG. 7A provides a side view of a truck comprising an electric rear axle.
- (9) FIG. 7B provides a side view of a conventional truck architecture.

DETAILED DESCRIPTION

(10) The invention is described more fully hereinafter with reference to the accompanying drawings, in which embodiments of the invention are shown. In the drawings, the absolute and relative sizes of systems, components, layers, and regions may be exaggerated for clarity. Embodiments may be described with reference to schematic and/or cross-section illustrations of possibly idealized embodiments and intermediate structures of the invention. In the description and drawings, like numbers refer to like elements throughout. Relative terms as well as derivatives thereof should be construed to refer to the orientation as then described or as shown in the drawing under discussion. These relative terms are for convenience of description and do not require that the system be constructed or operated in a particular orientation unless stated otherwise.

(11) FIG. 1 shows a top view of a truck **10** with a rear axle **400** with integrated electric motor according to a first embodiment. The rear axle **400** comprises a motor housing **440** extending laterally between rear wheels **11** of the truck **10**. As shown in FIG. 6, the rear axle **400** comprises an electric motor **410** and power transmission **420**, coaxially aligned in said motor housing **440** and arranged for driving the rear wheels **11**. For example, the rear axle can comprise two electric motors **410** and corresponding power transmissions **420**, each arranged for driving the rear wheels **11** on one side of the truck **10**, whereas the electric motors **410** are oriented back to back. This setup allows independent (electronic) control of the torque to the rear wheels **11** on each side of the truck **10**. Alternatively, a single electric motor **410** and power transmission **420** can be used to simultaneously drive the rear wheels **11** on both sides of the truck **10**, e.g. by having a hollow shaft electric motor and power transmission **420**. In case of a single electric motor **410**, the corresponding power transmission **420** may comprise a controllable differential drive or planetary gear system to mechanically divide the drive torque between the rear wheels **11** on each side.

(12) Turning back to FIG. 1, the truck **10** further comprises a suspension system **200**, for suspending the rear axle **400** to a chassis **15**. The suspension system **200** at least comprises a pair of springs **210** and shock absorbers **220**, and an anti-roll stabilizer **270** and a guiding rod linkage **280** for allowing movement in a vertical direction Z relative to the chassis **15**. In more detail with reference to FIG. 4, the motor housing **440** is limited by a minimum vertical ground clearance DH between the road surface **5** and a bottom surface of the motor housing **440**. The minimum vertical ground clearance DH may be at least fifteen centimeters. Preferably, the minimum vertical ground clearance is between fifteen and 25 centimeters.

(13) FIG. 1 shows that the motor housing **440** comprises a reinforced central top section **444** providing a guiding rod mount **285** for pivotally mounting the guiding rod linkage **280** to the rear axle **400**. The motor housing **440** further comprises suspension yokes **450** integrated into the housing. The suspension yokes **450** extend on either side of the motor housing **440** near both axial ends, and provide spring mounts **215** and shock absorber mounts **225** for mounting the opposed pair of springs **210** and shock absorbers **220**. As can be seen in the figure, a rear section **490** is provided on the motor housing **440**, provided with connectors **480** to provide input ports to the electric motor.

(14) Referring to FIG. 2 a further embodiment of a truck **10** is depicted with one or more high voltage cables **300**. Preferably, the high voltage cables **300** extend from chassis mount **510** in a backward (−X) direction along a longitudinal member **15** of the chassis, and approach the connectors **480** along the vertical (−Z) direction. The benefit of this configuration, is that the length of the power cables **300** can be made as long as possible within the given length of the truck **10**, which reduces the movement amplitude per centimeter of cable when the E-axle **400** moves

relative to the chassis **100**. By having the high voltage cables **300** extend along a longitudinal member **15** of the chassis **100**, e.g. on an inside surface of the longitudinal member **110**, the power cables can be shielded inside the chassis **100** against dirt, damage, and electromagnetic emission. (15) Preferably, the rear section outer surface **490** is easily accessible from the bottom of the truck, e.g. for service or maintenance. Preferably, the rear section outer surface **490** is a surface on a rear half, more preferably a rear bottom quadrant of the E-axle.

(16) Most preferably, the connectors are provided in an orientation to guide one or more high voltage power cables **300** adjacent the rear section outer surface **490** in a vertical upward direction +Z. For example, the connectors can be provided on a rear section outer surface **490** which is substantially flat and smooth, so that power cables **300** can closely be guided upward along the E-axle **400**. The connectors on the rear section outer surface **490** can e.g. be straight connectors oriented outward, while the connectors **480** on the power cables **300** are right-angle connectors **480**, or vice versa. Alternatively, the rear section outer surface **490** may for example be oriented relatively upward or perpendicular to the outer surface of the E-axle, so that straight connectors can be used on the rear section outer surface **490** and on the power cables **300**.

(17) FIG. 3 provides an embodiment of a support beam **500**, comprising bending stiffness elements extending along the cable path. In the embodiment of FIG. 3, three typical areas can be distinguished along the cable path **900**: (i) Starting at the chassis mount **510**, the support beam comprises a chassis end section, which extends horizontally in the -X direction from the chassis mount **510**. As the distance to the chassis mount **510** increases, the bending stiffness of the support beam **500** increases as well. (ii) Next, the support beam **500** transitions into a middle section, in which the bending stiffness of the support beam **500** reaches a maximum. (iii) Going towards the E-axle mount **520** provided at a rear section outer surface of the E-axle, an end section approaches the E-axle mount **520** vertically in the -Z direction along the cable path. Similar to the chassis end section, the bending stiffness of the E-axle end section is lower towards the E-axle mount **520** and increases as the distance thereto is increased. The cable may be connected to the support beam **500** by lateral mounting elements **590**.

(18) Additionally, in the configuration as shown in FIG. 3, the vertical orientation of the E-axle end section increases the (vertical) bending stiffness of that section of the support beam **500**. This may allow the absolute length of the E-axle end section to be longer than the chassis end section. The stiffness of the support beam **500** can be relatively less sensitive to the distance between the chassis mount **510** and E-axle mount **520** in the vertical direction Z, than to the distance between the chassis mount **510** and E-axle mount **520** in the horizontal direction X. By having a support beam **500** that approaches the E-axle mount **520** vertically in the -Z direction along cable path **900**, a longer length of power cable **400** can be supported by a less sensitive support beam **500**, which improves spreading the total movement of the power cable **400** equally along the cable path **900**.

(19) Referring to FIG. 4, the suspension yokes **450** further provide anti-roll mounts **275** on a bottom section of the motor housing **440** near both axial ends to mount the anti-roll stabilizer **270** to the rear axle **400**.

(20) Back to FIG. 1, on each axial end of the motor housing **440** the anti-roll mount **275**, the opposing pair of spring mounts **215** and the shock absorber mount **225** lie on a common plane **808** perpendicular to a centerline **101** of the rear wheels **11** of the truck **10**.

(21) The potential benefit of having the mounts **275**, **215**, **225** on a common plane **808** is, that reaction forces on the motor housing **440** caused by interaction of the rear axle **400** with the suspension system **300** are concentrated in a single plane **808**, which with topology optimization, may lead to significant weight reduction of the rear axle **400**.

(22) Preferably, the suspension yoke **450** is coaligned with plane **808** and designed to bear all corresponding suspension loads on the motor housing **440**, while other parts of the motor housing are largely unloaded and can therefore be relatively thin-walled or light-weight. A rear axle **400** with integrated electric motor **410** and power transmission **420** could have comparable weight as a

conventional driven axle.

(23) The compact size of the electric motor **410** and transmission **420** allows placing them in between the wheels, coaxial with the centerline **101** of the rear wheels **11**. This eliminates the need for an angled drive mechanism, and therefore reduces the total weight of the rear axle **400**. It is a challenge, however, to integrate the electric rear axle **400** into a conventional truck architecture without significantly changing the dynamic properties of the truck **10**.

(24) The current solution proposes to integrate carrying and driving functions into the rear axle **400**. Driving functions are e.g. provided by the driveshafts, the wheel hubs with their bearings, the wheels and tires. The carrying functions are for example provided by the motor housing **440**, where on conventional trucks a rigid driven axle is used. The suspension system **200** with springs **210** and shock absorbers **220**, together with the guiding rod linkage **280** and anti-roll stabilizer **270** are other examples of carrying functions. The rear axle assembly is typically fitted with brakes between the driving and carrying components.

(25) To reduce the total weight of the rear axle **400**, integration of the suspension yokes **450** that support the springs **210** into the motor housing **440** can provide a considerable weight reduction, saving the weight connections with clamping plates and U-bolts which are typically used in a conventional driven axle. Integration of the guiding rod mount **285** into the motor housing **440** may bring a weight reduction as well. The shape of the proposed motor housing **440** provides an opportunity to save weight in the mounting of the brakes as well. A conservative estimation indicates a weight saving of at least fifty kilograms with respect to a conventional driven axle with the measures above.

(26) FIG. **4** provides an isometric view of a further embodiment of the truck **10**, wherein the anti-roll stabilizer **270** comprises a torsion shaft **271** which is mounted to the chassis (not shown) while allowing an axial rotation at an offset to the rear axle **400**. The torsion shaft **271** has a pair of linkage plates **272**, **273** on both ends extending towards the anti-roll mounts **275**. In the anti-roll stabilizer configuration as described, a roll movement of the truck **10**, causing a difference in a vertical displacement of the anti-roll mounts **275** on each axial end of the motor housing **440** relative to the chassis **15**, is counteracted by a torsional stiffness of the torsion shaft **271**.

(27) Preferably, the linkage plates **272**, **273** are rigidly connected to the ends of the torsion shaft **271** and extend perpendicularly to the axis of rotation **303** of the torsion shaft **271** towards the anti-roll mounts **275**. Preferably, the linkage plates **272**, **273** have equal length, so that the torsion shaft **271** is mounted parallel to the rear axle **400**, such that the rear axle **400** is suspended to the chassis as part of a parallelogram mechanism comprising the anti-roll stabilizer and the guiding rod linkage, allowing a movement of the rear axle **400** in vertical direction Z, while other degrees of freedom of the rear axle **400** are constrained.

(28) The anti-roll stabilizer may comprise e.g. planar bearings, roller bearings or ball bearings on the ends of the torsion shaft **271** to mount the torsion shaft **271** to the chassis **15** while allowing axial rotation. Alternatively, the bearings can be mounted on a central section of the torsion shaft **271** between the linkage plates **272**, **273** to mount the torsion shaft **271** to the chassis **15** while allowing axial rotation.

(29) Preferably, the linkage plates **272**, **273** are pivotally mounted to the anti-roll mounts **275** on the motor housing **440**. For example, the connection between the linkage plates **272**, **273** and the anti-roll mounts may comprise planar bearings, ball bearings, or roller bearings.

(30) Alternatively, the linkage plates **272**, **273** can be rigidly connected to the motor housing **440** or may be an integral part of the suspension yokes **450**, or the torsion shaft **271** or entire anti-roll stabilizer **270** can be mounted at a different location or orientation to create a suspension system with alternative kinematics and dynamics. However, the reaction forces on the motor housing **440** caused by interaction of the rear axle **400** with the suspension system **300** may still be concentrated in a single plane **808**, as long as the anti-roll mount, the opposing pair of spring mounts **215** and the shock absorber mount **225** lie on a common plane **808** perpendicular to the centerline **101** of the

rear wheels **11** of the truck **10**.

(31) FIG. 5 provides an isometric view of another further embodiment of the truck **10**, wherein the guiding rod linkage **280** comprises a pair of guiding rods **281**, **282**. The pair of guiding rods **281**, **282** are pivotally mounted to the chassis (not shown) at an offset to the rear axle **400**, and extend towards the reinforced central top section **444**. Preferably, the guiding rods **281**, **282** have an equal length. The pair of guiding rods **281**, **282** can for example be mounted between the rear axle **400** and the chassis, such that the rear axle **400** is suspended to the chassis as part of a parallelogram mechanism comprising the guiding rod linkage and the anti-roll stabilizer, allowing a movement of the rear axle **400** in vertical direction Z, while other degrees of freedom of the rear axle **400** are constrained.

(32) For example, the pair of guiding rods **281**, **282** may form a V-shape, with the ends of the guiding rods that are mounted to the chassis spread along a common axis of rotation **202**, and the ends of the guiding rods that are mounted to the rear axle **400** converging toward the guiding rod mount **285**. Alternatively, the pair of guiding rods **281**, **282** can have a V-shape of which the converging end is mounted to the chassis, and the other ends are mounted to the motor housing **440**, near the axial ends, or to the suspension yokes **450**.

(33) In some preferred embodiments, the diverging ends of the V-shaped pair of guiding rods **281**, **282** is mounted to the suspension yokes together with the anti-roll mount, the opposing pair of spring mounts **215** and the shock absorber mount **225** on a common plane **808** perpendicular to a centerline **101** of the rear wheels **11** of the truck **10**. This may provide optimal distribution of reaction forces on the rear axle **400**, caused by interaction with the suspension system while the truck is driving, which with topology optimization, may lead to further weight reduction of the rear axle **400**.

(34) In a preferred embodiment, the reinforced central top section **444** of the motor housing **440** comprises a number of ribs converging toward the guiding rod mount **285** and having a rib height that is highest near the guiding rod mount **285** and that decreases away from the guiding rod mount **285**. Preferably, the rib height is aligned with the vertical direction Z, such that the vertical bending stiffness of the reinforced central top section **444** is increased. By having a number of ribs converge toward the guiding rod mount **285**, this can provide the guiding rod mount **285** with reinforced stiffness in all degrees of freedom. The ribs can e.g. be an integral part of the motor housing or be separately assembled parts.

(35) Alternatively, the reinforced central top section **444** is provided by the motor housing **440** having a larger wall thickness near the guiding rod mount **285**. However, compared to the embodiment with ribs, this may lead to a higher overall weight of the motor housing **440**. Alternatively, the reinforced central top section **444** can be provided by the motor housing **440** having an internal wall structure supporting the central top section. For example, the motor housing **440** may comprise a central wall that axially splits the rear axle into two opposite compartments, each housing components for driving wheels on one side of the truck **10**. Accordingly, the central wall may serve as a support wall to reinforce the central top section **444**.

(36) In yet further embodiments, housing extensions **470** are coaxially connected to both axial ends of the motor housing **440** for mounting rear wheel brakes **600**. The housing extensions **470** have a proximal base part that is axially connected to the motor housing **440**, and have a distal part extending away from the motor housing **440** toward the rear wheels **11** with a diameter smaller than the proximal base part. For example, the housing extensions **470** can be conical, horn shaped, or comprise a stepped outer surface.

(37) Alternatively, the housing extensions **470** are brackets connecting the rear wheel brakes **600** to the motor housing **440**. Alternatively, the housing extensions **470** can be an integral part of the motor housing **440**, or the motor housing **440** can provide surfaces for directly mounting the rear wheel brakes **600**. By integrating the suspension yokes **450** into the rear axle **400** the brake cylinders **610** can be moved to a lower position compared to a conventional truck, e.g. to a position

below the springs **210** as shown in FIG. **6**. This, in turn, enables locating the springs **210** closer to the centerline **101** of the rear axle, which reduces the length of the suspension yokes **450** and chassis **15**, and thus the total weight of the truck **10**. Having a short length of suspension yokes **450** in relation to the other dimensions of the rear axle **400** makes it feasible to integrate the suspension yokes **450** with the motor housing **440**, e.g. manufactured as a cast component.

(38) In some preferred embodiments, the motor housing **440** comprises a rear section outer surface **490** provided with connectors **480**, to provide input ports to the electric motor (not shown). Preferably, the rear section outer surface **490** is easily accessible from the bottom of the truck, e.g. for service or maintenance. Preferably, the rear section outer surface **490** is a surface on a rear half, more preferably a rear bottom quadrant of the motor housing **440**. In case of more than one connector, these are preferably provided parallel to the centerline **101** of the rear wheels, to have equal bending behavior of the cables to be connected to the connectors **480** through a mating cable connector **481**.

(39) Most preferably, the connectors **480** are provided in an orientation to guide one or more high voltage power cables **300** adjacent the rear section outer surface **490** in a vertical upward direction +Z. For example, the connectors **480** can be provided on a rear section outer surface **490** which is substantially flat and smooth, so that power cables **300** can closely be guided upward along the motor housing **440**. The connectors **480** on the rear section outer surface **490** can e.g. be straight connectors **480** oriented outward, while the cable connectors **481** of the power cables **300** are right-angle connectors, or vice versa. Alternatively, the rear section outer surface **490** may for example be oriented relatively upward or perpendicular to the outer surface of the motor housing **440**, so that straight connectors **480** can be used on the rear section outer surface **490** and on the power cables **300**.

(40) FIG. **6** provides an exploded view of yet another or further embodiment of the truck **10**, wherein the motor housing **440** is additionally arranged for internally carrying a gearbox and/or inverter **430** for powering different groups of coils of the electric motor **410**. Preferably, the gearbox and/or inverter **430** is coaxially aligned with the electric motor **410** and the power transmission **420**. For example, the gearbox and/or inverter **430** can be placed behind the electric motor **410** relative to the power transmission **420**, in a central section of the motor housing **440**.

(41) The electric motor **410** and power transmission **420** fit into the cavity inside the motor housing **440**. Considering the potential weight savings that could be realized on the rear axle assembly, there may be a weight budget to integrate a gearbox and/or inverter **430** into the rear axle **400** as well. This could bring advantages in the cable connections from the inverter **430** to the electric motor **410**. The inverter **430** is connected to the electric motor **410** by multiple cables to power different groups of coils. By having an inverter **430** placed close to the electric motor **410** the cables can be short, and since they are located inside the motor housing **440**, problems with electromagnetic radiation due to the alternating current can be limited by applying EMC shielding measures to the motor housing **440**.

(42) In a preferred embodiment, the motor housing **440** is made of an electrically conductive material to electromagnetically shield at least the electric motor **410**. In this way, the motor housing **440** can form a Faraday cage around the internally carried electric components to reduce emission of and susceptibility to electromagnetic radiation, which may be required for performance and reliability of the truck, or to comply to industry specific technical regulations. For example, the motor housing **440** can be made of cast iron, steel, or an aluminum alloy.

(43) Alternatively, the motor housing **440** is made of an electrically non-conductive material, such as a plastic, but has an electrically conductive element adjacent the inner or outer surface of the motor housing **440**, such as a conductive paint or a separate conductive mesh structure forming a Faraday cage around the internally carried components.

(44) Preferably, the motor housing **440** is made of a thermally conductive material with a thermal conductivity of at least ten Watt per meter Kelvin to passively cool e.g. the electric motor **410**. For

example, the thermal conductivity of the motor housing **440** can be between ten and seventy Watt per meter Kelvin for steel motor housings **440**, between thirty and one hundred Watt per meter Kelvin for (cast) Iron motor housings **440**, or between seventy and two hundred and forty Watt per meter Kelvin for Aluminum (alloy) motor housings **440**.

(45) In other or further preferred embodiments, the rear axle **400** is provided with an internal cooling circuit for circulating coolant fluid through the motor housing **440** and one or more coolant hose connectors on the rear section outer surface **490** for receiving one or more coolant hoses **350**, to actively cool at least the electric motor **410**. For example, the motor housing **440** may comprise internal walls adjacent the internally carried components, such as the electric motor **410**.

Accordingly, the internal walls can have channels for circulating coolant fluid through the motor housing **440** close to the source of heat.

(46) Alternatively or additionally, the external walls of the motor housing **440** may comprise channels for coolant fluid. Alternatively, a separate element comprising an internal cooling circuit, such as a heat sink or other type of heat exchanger, can be built into or onto the motor housing.

(47) Alternatively, the motor housing may comprise sealed compartments having a defined inlet and outlet, each compartment housing a component such as an electric motor or a power transmission, and being supplied with a flow of coolant fluid.

(48) Preferably, the coolant hoses are fluidly connected to an external pump and reservoir containing coolant fluid, which is pumped to and from the rear axle **400**. The rear axle cooling system may be part of a larger truck cooling system, cooling other parts on the truck **100** as well to reduce the total weight of the truck **10**.

(49) Alternatively, the rear axle **400** may comprise its own independent cooling system, e.g. if the cooling capacity required for cooling the rear axle **400** cannot be matched with that of other parts of the truck **10**.

(50) Preferably, the coolant fluid is additionally employed as lubricant for lubricating rotating parts inside the motor housing **440** such as the power transmission **420**, to avoid requiring separated flows of lubricant and coolant fluid, and thus reduce the total weight of the truck **10**.

(51) FIG. 7A provides a side view of a truck **10** comprising an electric rear axle **400**. In comparison, FIG. 7B provides a side view of a conventional truck **70** architecture, comprising an internal combustion engine **71**, a power transmission **72** and propshaft **73**, driving the rear wheels **11** of the truck through a conventional driven axle **78**. The conventional driven axle **78** and electric rear axle **400** can be mounted to the chassis **15** by the same guiding rod linkage **280**, and at the same chassis mount **286** and guiding rod mount **285**. In both architectures, the anti-roll stabilizer **270** can have the same dimensions, orientation, and mounts **275** and **276**. This facilitates integration of electric rear axle **400** into conventional truck architecture **70**. Because of the obsolescence of a moving propshaft **73**, in an electric rear axle **400** (see FIG. 5) the torsion shaft **271** of the anti-roll stabilizer can be placed at a more advantageous location between chassis mounts **276** compared to the conventional driven axle **78**, which may benefit manufacturability and serviceability.

(52) For suspension kinematics and secondary torques due to roll of the truck **10**, it can be an advantage to have a short distance in the X direction between the anti-roll mounts **275** and the centerline **101** of the rear wheels **11**. In the conventional driven axle **78** the torsion shaft is preferably located between or near mounts **275**. When the same anti-roll stabilizer layout would be integrated into an electric rear axle **400**, the relatively bulkier shape of the motor housing **440** would force the torsion shaft **271** forward in X-direction. A more forward mounted torsion shaft implies a larger distance in X-direction between the anti-roll mounts **275** and the centerline **101** of the rear axle **400** as well, with negative effects on kinematics and secondary torque on the axle.

(53) The inventive solution is to flip the anti-roll stabilizer **270** over one hundred and eighty degrees along the Z-direction before mounting it to the rear axle **400**. As shown in FIG. 5, this implies that the torsion shaft **271** is mounted to the chassis **15** and the linkage plates **272**, **273** are

mounted to the anti-roll mounts 275 on the rear axle 400. The function of the anti-roll stabilizer 270 is thereby kept similar to one on a conventional driven axle 78.

Claims

1. A truck, comprising: a rear axle, comprising a motor housing extending laterally between rear wheels of the truck, and an electric motor and power transmission coaxially aligned in said motor housing and arranged for driving the rear wheels; and a suspension system, for suspending the rear axle to a chassis of the truck, said suspension system at least comprising a pair of springs and shock absorbers, and an anti-roll stabilizer and a guiding rod linkage for allowing movement in a vertical direction relative to the chassis; wherein the motor housing is limited by a minimum vertical ground clearance between the road surface and a bottom surface of the motor housing; wherein the motor housing comprises a reinforced central top section providing a guiding rod mount for pivotally mounting the guiding rod linkage to the rear axle; wherein the motor housing further comprises suspension yokes integrated into the housing and extending on either side of the motor housing near both axial ends, said suspension yokes providing spring mounts and shock absorber mounts for mounting said opposed pair of springs and shock absorbers, wherein the motor housing comprises a rear section outer surface provided with connectors, to provide input ports to the electric motor and wherein said connectors are provided in an orientation to guide one or more high voltage power cables adjacent the rear section outer surface in a vertical upward direction.
2. The truck according to claim 1, further comprising a support beam, forming a mechanical support structure to lead the power cable along a cable path extending between a chassis mount and an E-axle mount.
3. The truck according to claim 2, wherein the support beam further comprises an electrically conductive core, mounted between the chassis mount and the E-axle mount, to make a ground connection between the E-axle and the chassis.
4. The truck according to claim 2, wherein the support beam is mounted to the conductive core along at least a part of the cable path.
5. The truck according to claim 2, wherein the support beam comprises one or more bending stiffness elements, forming a stiffness of the support beam that is highest in a middle section and that decreases towards the chassis mount and the E-axle mount, evenly distributing the bending stress in the support beam along the cable path when the E-axle moves relative to the chassis, to prevent local fatigue failure of the power cable.
6. The truck according to claim 5, wherein the bending stiffness elements comprise vertically extending flanges to prevent the support beam from deflecting in a lateral direction.
7. The truck according to claim 1, wherein said suspension yokes further provide anti-roll mounts to mount the anti-roll stabilizer to the rear axle and wherein the anti-roll stabilizer comprises a torsion shaft, said torsion shaft being mounted to the chassis while allowing an axial rotation at an offset to the rear axle, and having a pair of linkage plates on both ends extending from the torsion shaft towards the anti-roll mounts, such that a roll movement of the truck, causing a difference in a vertical displacement of the anti-roll mounts on each axial end of the motor housing relative to the chassis, is counteracted by a torsional stiffness of the torsion shaft.
8. The truck according to claim 1, wherein the guiding rod linkage comprises a pair of guiding rods, said pair of guiding rods being pivotally mounted to the chassis at an offset to the rear axle, and extending towards the reinforced central top section.
9. The truck according to claim 1, wherein the reinforced central top section of the motor housing comprises a number of ribs converging toward the guiding rod mount and having a rib height that is highest near the guiding rod mount and that decreases away from the guiding rod mount.
10. The truck according to claim 1, wherein housing extensions are coaxially connected to both axial ends of the motor housing for mounting rear wheel brakes, said housing extensions having a

proximal base part that is axially connected to the motor housing, and having a distal part extending away from the motor housing toward the rear wheels with a diameter smaller than the proximal base part.

11. The truck according to claim 1, wherein the motor housing is additionally arranged for internally carrying a gearbox and/or inverter for powering different groups of coils of the electric motor.

12. The truck according to claim 1, wherein the motor housing is made of an electrically conductive material to electromagnetically shield at least the electric motor.

13. The truck according to claim 1, wherein the motor housing is made of a thermally conductive material with a thermal conductivity of at least ten Watt per meter Kelvin.

14. The truck according to claim 1, wherein the rear axle is provided with an internal cooling circuit for circulating coolant fluid through the motor housing and one or more coolant hose connectors on the rear section outer surface for receiving one or more coolant hoses, to actively cool at least the electric motor.

15. The truck according to claim 14, wherein the coolant fluid is additionally employed as lubricant for lubricating rotating parts inside the motor housing.
